

Dopamine: Drug information - UpToDate

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For additional information see "Dopamine: Patient drug information" and "Dopamine: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Pharmacologic Category

- Adrenergic Agonist Agent;
- Inotrope

Dosing: Adult

Dosage guidance:

Dosing: Hemodynamic effects of dopamine are dose dependent; however, there is overlap of clinical effects between dosing ranges (Ref):

Low dose (<5 mcg/kg/minute): Augments renal dopamine receptors.

Intermediate dose (5 to 10 mcg/kg/minute): Dopamine and beta-adrenergic effects predominate, resulting in increased heart rate, cardiac contractility, and cardiac output.

High dose (>10 mcg/kg/minute): Alpha-adrenergic effects begin to predominate, resulting in vasoconstriction and increased blood pressure.

Atrioventricular block, symptomatic and/or hemodynamically unstable

Atrioventricular block, symptomatic and/or hemodynamically unstable (off-label use):

Note: Should only be used as a temporizing measure until a permanent pacemaker can be placed, if clinically indicated. In most cases, pharmacologic agents will not be effective and temporary transcutaneous or transvenous pacing is necessary as a bridge to permanent pacemaker placement (Ref).

Continuous infusion: IV: Initial: 5 mcg/kg/minute; increase dose every 2 minutes until desired effect; maximum dose: 20 mcg/kg/minute (Ref).

Bradycardia, symptomatic and/or hemodynamically unstable

Bradycardia, symptomatic and/or hemodynamically unstable (unresponsive to atropine) (off-label use):

Continuous infusion: IV: Initial: 5 mcg/kg/minute; increase dose every 2 minutes until desired effect; maximum dose: 20 mcg/kg/minute (Ref).

Hypotension or shock

Hypotension or shock:

Cardiogenic shock (alternative agent):

Note: Typically, not the preferred initial agent in cardiogenic shock due to increased risk for arrhythmias and possibly mortality; consider other inotropic and/or vasopressor options (Ref). Optimal goal of therapy is not well established, but typically titrate to maintain end-organ perfusion (Ref).

Continuous infusion: IV: Usual dosage range: 2 to 20 mcg/kg/minute; titrate based on clinical end point (eg, end-organ perfusion) (Ref).

Septic shock and other vasodilatory shock states (alternative agent):

Note: Not recommended for septic shock; may consider as an alternative to norepinephrine in patients with bradycardia who have a low risk of tachyarrhythmias or when norepinephrine is not available (Ref).

Continuous infusion: IV: Initial: 2 to 5 mcg/kg/minute; titrate to clinical endpoint (eg, BP or other perfusion measure) up to a dose of 20 mcg/kg/minute (Ref).

Post-cardiac arrest shock (alternative agent):

Note: Typically, not the preferred initial agent in post-cardiac arrest shock due to risk of tachyarrhythmias; consider other inotropic and/or vasopressor options (Ref). Optimal goal of therapy is not well established, but typically titrate to MAP >65 mm Hg (Ref).

Continuous infusion: IV: Usual dosage range: 5 to 20 mcg/kg/minute; titrate based on clinical end points (eg, MAP, end-organ perfusion) (Ref).

Inotropic support

Inotropic support:

Note: May consider in patients with severe systolic dysfunction (eg, ejection fraction $\leq 40\%$) with decreased end-organ perfusion (Ref).

Continuous infusion: IV: 5 to 15 mcg/kg/minute; doses at lower end of this range are preferred as inotropic actions predominate at lower doses and vasoconstrictive actions predominate at higher doses (Ref).

Dosing: Kidney Impairment: Adult

No dosage adjustment necessary (cleared by nonrenal mechanisms and titrated to effect) (Nemecek 2019).

Dosing: Obesity: Adult

The recommendations for dosing in patients with obesity are based upon the best available evidence and clinical expertise. Senior Editorial Team: Jeffrey F. Barletta, PharmD, FCCM; Manjunath P. Pai, PharmD, FCP; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC.

Class 1, 2, and 3 obesity (BMI ≥ 30 kg/m²):

Continuous infusion: IV: Use **ideal body weight** for initial dose calculations, then titrate to hemodynamic effect and clinical response (Ref). During therapy, clinicians should **not** change dosing weight from one weight metric to another (ie, ideal body weight to/from actual body weight) (Ref). Refer to adult dosing for indication specific doses.

Rationale for recommendations:

There is a paucity of studies evaluating the influence of obesity on dopamine dosing or pharmacokinetics. One observational study evaluating patients with obesity and overall mortality showed a trend in lower dopamine doses at 6 hours with increasing obesity (Ref). Other observational studies, including phenylephrine, epinephrine and norepinephrine, suggest non-weight-based dosing strategies may result in lower overall cumulative dose requirements and increased drug exposure to second line agents in some patients and may not be advantageous in time to achieving hemodynamic

stability (Ref). However, it is difficult to show outcome differences between weight-based and non-weight-based dosing because of dose titration to target BP, particularly in the context of retrospective studies. Furthermore, there is substantial variability in response in critically ill patients, irrespective of weight. Due to the short onset of action and small volume of distribution, rapid titration to clinical effect after initial dosing is possible (Ref).

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

(For additional information see "Dopamine: Pediatric drug information")

Hemodynamic support

Hemodynamic support: Infants, Children, and Adolescents: Continuous IV or intraosseous infusion: 2 to 20 mcg/kg/minute; titrate gradually by 5 to 10 mcg/kg/minute increments until optimal response is obtained (Ref). **Note:** Dopamine has a dose-dependent effect; doses at lower end of this range are preferred, as inotropic actions predominate at lower doses and vasoconstrictive actions predominate at higher doses (Ref).

Adverse Reactions

The following adverse drug reactions are derived from product labeling unless otherwise specified.

Postmarketing:

Cardiovascular: Angina pectoris, atrial fibrillation, bradycardia, cardiac conduction disorder, ectopic beats, hypertension, hypotension, palpitations, tachycardia, vasoconstriction, ventricular arrhythmia, widened QRS complex on ECG

Dermatologic: Peripheral gangrene (with prolonged or high dose, can occur with low doses with concomitant occlusive vascular disease), piloerection

Gastrointestinal: Nausea, vomiting

Genitourinary: Azotemia

Nervous system: Anxiety, headache

Respiratory: Dyspnea

Contraindications

Pheochromocytoma.

Warnings/Precautions

Concerns related to adverse effects:

- Arrhythmias: May increase the risk of tachycardia and other tachyarrhythmias including ventricular arrhythmias (Tisdale 1995). In heart transplant candidates, institute appropriate measures to protect patient against risks of sudden cardiac death (Young 2000).
- Extravasation: Vesicant; infuse into a large vein if possible. Avoid infusion into leg veins.

Disease-related concerns:

- Cardiovascular disease: Use with caution in patients with cardiovascular disease, cardiac arrhythmias and/or occlusive vascular disease.
- Active myocardial ischemia/post-myocardial infarction: Use with caution in patients with active myocardial ischemia or recent myocardial infarction; may increase myocardial oxygen consumption.
- Electrolyte imbalance: Correct electrolyte disturbances, especially hypokalemia or hypomagnesemia, prior to use and throughout therapy to minimize the risk of arrhythmias (ACC/AHA/ESC [Zipes 2006]; Tisdale 1995).

- Shock: The use of dopamine in adult patients with shock has demonstrated a higher incidence of adverse events (eg, tachyarrhythmias) compared to alternative vasopressors (De Backer 2010). Higher 28-day mortality was also seen in patients with septic shock with the use of dopamine as compared to norepinephrine (De Backer 2012; Vasu 2012).

Dosage form specific issues:

- Sodium metabisulfite: Product may contain sodium metabisulfite.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution, Intravenous, as hydrochloride:

Generic: 0.8 mg/mL (250 mL, 500 mL); 1.6 mg/mL (250 mL, 500 mL); 3.2 mg/mL (250 mL); 40 mg/mL (5 mL, 10 mL); 3.2 mg/mL (250 mL [DSC])

Solution, Intravenous, as hydrochloride [preservative free]:

Generic: 1.6 mg/mL (250 mL [DSC], 500 mL); 40 mg/mL (5 mL, 10 mL); 1.6 mg/mL (250 mL, 500 mL); 3.2 mg/mL (250 mL)

Generic Equivalent Available: US

Yes

Pricing: US

Solution (DOPamine HCl Intravenous)

40 mg/mL (per mL): \$0.70 - \$1.06

Solution (DOPamine-Dextrose Intravenous)

0.8 mg/mL 5% (per mL): \$0.06

1.6 mg/mL 5% (per mL): \$0.07 - \$0.13

3.2 mg/mL 5% (per mL): \$0.09 - \$0.12

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Intravenous, as hydrochloride:

Generic: 0.8 mg/mL (250 mL); 1.6 mg/mL (250 mL, 500 mL); 3.2 mg/mL (250 mL, 500 mL)

Administration: Adult

IV: Administer as a continuous infusion via an infusion pump. Central line administration is preferred; extravasation may cause severe ischemic necrosis. If central line is not available, may administer for a short duration (<72 hours) through a peripheral IV catheter placed in a large vein at a proximal site (eg, in or proximal to antecubital fossa). Frequent monitoring of the IV catheter site is

recommended to rapidly identify extravasation (Ref). Refer to institutional policies and procedures; catheter placement/size and vasopressor concentration may vary depending on institution.

Vesicant; ensure proper needle or catheter placement prior to and during infusion; avoid extravasation.

Extravasation management: If extravasation occurs, stop infusion immediately; leave cannula/needle in place temporarily but do **NOT** flush the line; gently aspirate extravasated solution, then remove needle/cannula; elevate extremity; apply dry, warm compresses; initiate phentolamine (or alternative) antidote (Ref).

Phentolamine: **SUBQ:** Dilute 5 to 10 mg in 10 mL 0.9% sodium chloride and administer into extravasation site as soon as possible after extravasation; if infiltrated catheter has not been removed, administer initial dose through the infiltrated catheter; may repeat in 60 minutes if patient remains symptomatic (Ref).

Alternatives to phentolamine:

Nitroglycerin topical 2% ointment: Apply a 1-inch strip to the site of ischemia to cover the affected areas; may repeat every 8 hours as necessary (Ref).

Terbutaline:

Large extravasations: **SUBQ:** Infiltrate affected extravasation area with terbutaline 1 mg using a solution of terbutaline 1 mg diluted in 10 mL 0.9% sodium chloride; may repeat dose after 15 minutes (Ref).

Small extravasations: **SUBQ:** Infiltrate affected extravasation area with 0.5 mg using a solution of terbutaline 1 mg diluted in 1 mL 0.9% sodium chloride; may repeat dose after 15 minutes (Ref).

Preparation for Administration: Adult

Conventional vials must be diluted prior to administration. Transfer contents of 1 or more vials by aseptic technique to either 250 or 500 mL of a compatible sterile IV solution. Premixed infusion solution does not require dilution prior to administration; do not add additional medications to premixed solution. Solution should be clear, practically colorless; do not administer if solution is darker or discolored.

Usual Infusion Concentrations: Adult

Note: Premixed solutions available.

IV infusion: 400 mg in 250 mL (concentration: 1600 mcg/mL) **or** 800 mg in 250 mL (concentration: 3200 mcg/mL) of D₅W or NS

Administration: Pediatric

Parenteral: Continuous IV infusion: Vials (concentrated solution) must be diluted prior to administration; premixed IV solutions (800 mcg/mL, 1,600 mcg/mL, 3,200 mcg/mL) are available. Administer as a continuous IV infusion with the use of an infusion pump or an intraosseous infusion until IV access can be obtained in pediatric patients (Ref). Administer into large vein to minimize the possibility of extravasation (central line administration); administration into an umbilical arterial catheter is **not** recommended (Ref). If central line is not available, may administer for a short duration through a peripheral IV catheter placed in a large vein or via intraosseous access using a more dilute solution or with a second carrier fluid; once central access is available, begin central line infusion and wait for pharmacologic effect prior to stopping peripheral administration (Ref). Frequent monitoring of the IV catheter site is recommended to rapidly identify extravasation (Ref). Refer to institutional policies and procedures; catheter placement/size and vasopressor concentration may vary depending on institution. Do not administer sodium bicarbonate (or any alkaline solution) through an IV line containing dopamine; inactivation of dopamine may occur (Ref). Avoid abrupt discontinuation; reduce infusion flow rate slowly.

Rate of infusion (mL/hour) = [dose (mcg/kg/minute) × weight (kg) × 60 minutes/hour] divided by concentration (mcg/mL)

Vesicant; ensure proper needle or catheter placement prior to and during infusion; avoid extravasation. If extravasation occurs, stop infusion immediately; leave cannula/needle in place temporarily but do **NOT** flush the line; gently aspirate extravasated solution, then remove needle/cannula unless needed for IV antidote; elevate extremity; apply dry, warm compresses. Initiate phentolamine (or alternative antidote) (Ref).

Preparation for Administration: Pediatric

Parenteral: Continuous IV infusion: Vials (concentrated solution) must be diluted prior to administration in a compatible solution (eg, NS, D5W, D5NS, D5¹/₂NS, LR, D5LR) to a maximum concentration of 3,200 mcg/mL; concentrations as high as 6,000 mcg/mL (6 mg/mL) have been reported (Ref); premixed IV solutions are available (800 mcg/mL, 1,600 mcg/mL, and 3,200 mcg/mL). Vermont Oxford Network recommends a standard concentration of 800 mcg/mL or 1,600 mcg/mL for neonates (Ref). In pediatric patients, ASHP recommends standard concentrations of 800 mcg/mL, 1,600 mcg/mL, and 3,200 mcg/mL (Ref).

Usual Infusion Concentrations: Pediatric

Note: Premixed solutions available.

IV infusion: 800 **mcg/mL**, 1,600 **mcg/mL**, or 3,200 **mcg/mL**.

Vesicant/Extravasation Risk

Vesicant

Storage/Stability

Store vials and premixed single-use containers at 20°C to 25°C (68°F to 77°F). Avoid excessive heat; brief exposure of premixed single-use containers of up to 40°C (104°F) does not adversely affect the product. Protect from freezing. Dopamine has been found to be stable for a minimum of 24 hours after dilution in a compatible solution. Avoid contact or simultaneous administration with alkalies (including sodium bicarbonate), oxidizing agents, or iron salts.

Use: Labeled Indications

Hypotension or shock: Treatment of severe hypotension or shock (eg, septic shock and other vasodilatory shock states, cardiogenic shock, decompensated heart failure, post-cardiac arrest) that persists during and after adequate fluid volume replacement.

Use: Off-Label: Adult

Atrioventricular block, symptomatic and/or hemodynamically unstable; Bradycardia, symptomatic and/or hemodynamically unstable (unresponsive to atropine)

Medication Safety Issues

Sound-alike/look-alike issues:

DOPamine may be confused with DOBUTamine, Dopram

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (inotropic medications, IV) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024).

Metabolism/Transport Effects

Substrate of COMT, OCT1, OCT2

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Alpha1-Blockers: May decrease therapeutic effects of Alpha-/Beta-Agonists. *Risk C: Monitor*

Atomoxetine: May increase hypertensive effects of Sympathomimetics. Atomoxetine may increase tachycardic effects of Sympathomimetics. *Risk C: Monitor*

Benzylpenicilloyl Polylysine: Coadministration of Alpha-/Beta-Agonists and Benzylpenicilloyl Polylysine may alter diagnostic results. Management: Consider delaying skin testing until alpha/beta agonists are no longer required and their effects have diminished, or use a histamine skin test as a positive control to assess a patient’s ability to mount a wheal and flare response. *Risk D: Consider Therapy Modification*

Bitter Orange: May increase adverse/toxic effects of Sympathomimetics. *Risk C: Monitor*

Bornaprine: Sympathomimetics may increase anticholinergic effects of Bornaprine. *Risk C: Monitor*

Bromocriptine: May increase hypertensive effects of Alpha-/Beta-Agonists. Management: Consider alternatives to this combination when possible. If combined, monitor for hypertension and tachycardia, and do not coadminister these agents for more than 10 days. *Risk D: Consider Therapy Modification*

Cannabinoid-Containing Products: May increase tachycardic effects of Sympathomimetics. *Risk C: Monitor*

Chloroprocaine (Systemic): May increase hypertensive effects of Alpha-/Beta-Agonists. *Risk C: Monitor*

CloZAPine: May decrease therapeutic effects of Alpha-/Beta-Agonists. *Risk C: Monitor*

Cocaine (Topical): May increase hypertensive effects of Sympathomimetics. Management: Consider alternatives to use of this combination when possible. Monitor closely for substantially increased blood pressure or heart rate and for any evidence of myocardial ischemia with concurrent use. *Risk D: Consider Therapy Modification*

COMT Inhibitors: May increase serum concentrations of COMT Substrates. *Risk C: Monitor*

Dihydralazine: Sympathomimetics may decrease therapeutic effects of Dihydralazine. *Risk C: Monitor*

Doxapram: Sympathomimetics may increase adverse/toxic effects of Doxapram. *Risk C: Monitor*

Doxofylline: Sympathomimetics may increase adverse/toxic effects of Doxofylline. *Risk C: Monitor*

Droxidopa: Sympathomimetics may increase hypertensive effects of Droxidopa. *Risk C: Monitor*

Ergot Derivatives (Vasoconstrictive CYP3A4 Substrates): May increase vasoconstricting effects of Alpha-/Beta-Agonists. *Risk X: Avoid*

Esketamine (Injection): May increase adverse/toxic effects of Sympathomimetics. Specifically, the risk for elevated heart rate, hypertension, and arrhythmias may be increased. *Risk C: Monitor*

Guanethidine: May increase hypertensive effects of Sympathomimetics. Guanethidine may increase arrhythmogenic effects of Sympathomimetics. *Risk C: Monitor*

Hexoprenaline: May increase adverse/toxic effects of Alpha-/Beta-Agonists. *Risk X: Avoid*

Hyaluronidase: May increase vasoconstricting effects of Alpha-/Beta-Agonists. Management: Do not use hyaluronidase to enhance the dispersion or absorption of alpha-/beta-agonists. Use of hyaluronidase for other purposes in patients receiving alpha-/beta-agonists may be considered as clinically indicated. *Risk D: Consider Therapy Modification*

Inhalational Anesthetics: May increase arrhythmogenic effects of DOPamine. *Risk C: Monitor*

Kratom: May increase adverse/toxic effects of Sympathomimetics. *Risk X: Avoid*

Landiolol: Sympathomimetics may decrease therapeutic effects of Landiolol. *Risk C: Monitor*

Linezolid: May increase hypertensive effects of Sympathomimetics. Management: Consider initial dose reductions of sympathomimetic agents, and closely monitor for enhanced blood pressure elevations, in patients receiving linezolid. *Risk D: Consider Therapy Modification*

Lisuride: May increase hypertensive effects of Alpha-/Beta-Agonists. Management: Consider alternatives when possible, as product labeling states the concomitant use of lisuride with alpha-/beta-agonists is not recommended. If combined, monitor for increases in blood pressure or hypertensive crises. *Risk D: Consider Therapy Modification*

Melperone: May decrease therapeutic effects of DOPamine. *Risk C: Monitor*

Monoamine Oxidase Inhibitors: May increase hypertensive effects of DOPamine. Management: Initiate dopamine at no greater than one-tenth (1/10) of the usual dose in patients who are taking (or have taken within the last 2 to 3 weeks) monoamine oxidase inhibitors. Monitor for an exaggerated hypertensive response to dopamine. *Risk D: Consider Therapy Modification*

Pergolide: May increase hypertensive effects of Alpha-/Beta-Agonists. *Risk C: Monitor*

Serotonin/Norepinephrine Reuptake Inhibitor: May increase tachycardic effects of Alpha-/Beta-Agonists. Serotonin/Norepinephrine Reuptake Inhibitor may increase vasopressor effects of Alpha-/Beta-Agonists. Management: If possible, avoid coadministration of direct-acting alpha-/beta-agonists and serotonin/norepinephrine reuptake inhibitors. If coadministered, monitor for increased sympathomimetic effects (eg, increased blood pressure, chest pain, headache). *Risk D: Consider Therapy Modification*

Solriamfetol: Sympathomimetics may increase hypertensive effects of Solriamfetol. Sympathomimetics may increase tachycardic effects of Solriamfetol. *Risk C: Monitor*

Sympathomimetics: May increase adverse/toxic effects of Sympathomimetics. *Risk C: Monitor*

Tedizolid: May increase adverse/toxic effects of Sympathomimetics. Specifically, the risk for increased blood pressure and heart rate may be increased. *Risk C: Monitor*

Thyroid Products: May increase adverse/toxic effects of Sympathomimetics. Specifically, the risk of hypertension, tachycardia, and coronary insufficiency may be increased in patients with coronary artery disease. *Risk C: Monitor*

Tricyclic Antidepressants: May increase vasopressor effects of Alpha-/Beta-Agonists. Management: Avoid, if possible, the use of alpha-/beta-agonists in patients receiving tricyclic antidepressants. If combined, monitor for evidence of increased pressor effects and consider reductions in initial dosages of the alpha-/beta-agonist. *Risk D: Consider Therapy Modification*

Vasopressin: Alpha-/Beta-Agonists (Direct-Acting) may increase hypertensive effects of Vasopressin. The effect of other hemodynamic parameters may also be enhanced. *Risk C: Monitor*

Pregnancy Considerations

Medications required for the treatment of critically ill pregnant patients should not be withheld due to concerns of fetal teratogenicity (ACOG 2019; AHA [Jeejeebhoy 2015]). Medications used for the treatment of cardiac arrest in pregnancy are the same as in the nonpregnant patients. Dopamine use during the postresuscitation phase may be considered; however, the effects of vasoactive medications on the fetus should also be considered. Doses and indications should follow current Advanced Cardiovascular Life Support guidelines (AHA [Jeejeebhoy 2015]).

Breastfeeding Considerations

It is not known if dopamine is present in breast milk.

Monitoring Parameters

Note: General monitoring parameters are below; individualize monitoring and consult institutional policies and procedures.

ECG; hemodynamic parameters as appropriate (eg, BP, CVP, heart rate, CI, PCWP, SVR, ScvO₂ or SvO₂); end-organ perfusion; infusion site for blanching/extravasation.

Mechanism of Action

Stimulates both adrenergic and dopaminergic receptors, lower doses are mainly dopaminergic stimulating and produce renal and mesenteric vasodilation, higher doses also are both dopaminergic and beta₁-adrenergic stimulating and produce cardiac stimulation and renal vasodilation; large doses stimulate alpha-adrenergic receptors

Pharmacokinetics (Adult Data Unless Noted)

Note: Children: Dopamine has exhibited nonlinear kinetics in children; with dose changes, may not achieve steady-state for ~1 hour rather than 20 minutes.

Onset of action: Adults: Within 5 minutes.

Duration: Adults: <10 minutes.

Metabolism: Renal, hepatic, plasma; 75% to inactive metabolites by monoamine oxidase and 25% to norepinephrine (active).

Half-life elimination: ~2 minutes.

Excretion: Urine (as metabolites).

Clearance: Neonates: Varies and appears to be age related; clearance is more prolonged with combined hepatic and kidney dysfunction.

Patient Counseling Points

(For additional information see "Dopamine: Patient drug information")

What is this drug used for?

- It is used to treat heart failure (weak heart).
- It is used to treat low blood pressure.
- It may be given to you for other reasons. Talk with the doctor.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

- Upset stomach or throwing up
- Headache
- Anxiety
- Goosebumps

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- High blood pressure like very bad headache or dizziness, passing out, or change in eyesight
- Shortness of breath
- Chest pain or pressure
- Fast, slow, or abnormal heartbeat
- Cool or pale arm or leg
- Change in skin color to black or purple
- Not able to pass urine or change in how much urine is passed

- Redness, burning, pain, swelling, blisters, skin sores, or leaking of fluid where the drug is going into your body
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Domine | Dopamin | Dopamine | Dopmin;
- (AR) Argentina: Dopamina duncan | Dopamina fabra | Dopamina northia | Dopamina surar pharma | Dopatropin | Hettytropin | Inotropin | Megadose;
- (AT) Austria: Dopamin | Dopamin ebewe | Dopamin Solvay;
- (AU) Australia: Dopamine | Dopamine junio;
- (BD) Bangladesh: Cardopa | Dopamin | Myomine;
- (BE) Belgium: Dynatra;
- (BG) Bulgaria: Dopamin | Dopamin Admeda | Dopmin;
- (BR) Brazil: Cloridrato de dopamina | Constriction | Dopabane | Dopacris | Dopamin | Dopamina | Dopamol | Inotropisa | Revimine | Revivan | Teudom;
- (CH) Switzerland: Dopamin braun | Dopamin fresenius | Dopamin Giuliani | Dopamin Sintetica;
- (CL) Chile: Dopamina | Dopamina clorhidrato;
- (CN) China: Dopamine;
- (CO) Colombia: Cardiopal | Dopamina | Dopamina clorhidrato | Doparis | Inotropisa | Myofast;
- (CR) Costa Rica: Dopamina clorhidrato;
- (CZ) Czech Republic: Dopamin | Dopamin Admeda;
- (DE) Germany: Dopamin | Dopamin natter | Dopamin Solvay;
- (DO) Dominican Republic: Dopamina;
- (EC) Ecuador: Dopamina;
- (EE) Estonia: Dopamin | Dopamina grifols | Dopamine | Dopamine wzf polfa | Dopar | Dopmin;
- (EG) Egypt: Dopamine | Dopasunny;
- (ES) Spain: Aprical dopamida | Dopamina fides | Dopamina grifols;
- (ET) Ethiopia: Dopadren | Dopamine;
- (FI) Finland: Abbodop | Dopmin;
- (FR) France: Dopamine aguettant | Dopamine lucien | Dopamine merck | Dopamine nativele | Dopamine PCH | Dopamine pierre fabre | Dopamine Renaudin;
- (GB) United Kingdom: Dopamine | Dopamine premix | Ims dopamine | Intropin | Intropin ahs;
- (GR) Greece: Dopamine hycrochloride | Giludop;
- (GT) Guatemala: Dopamina clorhidrato;
- (HU) Hungary: Dopamin Giuliani | Dopamin Solvay;

- (ID) Indonesia: Cetadop | Dopac | Dopamin | Dopamin Giulini | Dopamine hcl abbott | Doperba | Indop | Oridop | Proinfark | Udopa;
- (IE) Ireland: Dopamine;
- (IN) India: Domin | Dopacef | Dopacin | Dopamine | Dopamine-p | Dopan | Dopinga | Dopress;
- (IT) Italy: Dopamina abbott | Dopamina cloridrato | Dopamina phd | Dopamina salif | Revivan;
- (JO) Jordan: Domine | Dopamine | Dopamine fresenius;
- (JP) Japan: Actopamin | Catabon | Catherine | Catherine amel | Corskool | Critpan | Dasomin | Dolpamil | Dominin | Donapan | Doncal | Dopa kit | Dopamine | Dopamine hydrochloride isei | Dopamine hydrochloride kobayashi | Dopamine Hydrochloride Mochida | Dopamine Hydrochloride Shionogi | Dopamine hydrochloride tsuruhara | Dopamine towa | Dopaminex | Doparalmin | Elebumin | Evetant hexal | Evetant towa | Gabans | Inovon | Kakodin | Martburn | Mitasmin | Predopa | Predopa Kyowa | Predopa merck hoei | Seminiet | Taiadopa | Tronjin | Tsurudopami | Yaelista | Yaelista mita;
- (KE) Kenya: Dopamine hcl fresenius;
- (KR) Korea, Republic of: Domine | Domins | Dopamine | Dopamine injection huons | Dopaminex | Dopan | Dopasin | Dopramin | Inophan | Tropin | Udopa;
- (KW) Kuwait: Dopamine | Intropin;
- (LB) Lebanon: Dopamine | Dopamine Renaudin;
- (LT) Lithuania: Dopamin | Dopamin Giulini | Dopmin;
- (LU) Luxembourg: Dopamin | Dynatra;
- (LV) Latvia: Dopamin | Dopmin;
- (MX) Mexico: Dopamina | Dopamina 3m | Dopamina exakta | Dopamina gi zafiro | Dopamina kendrick | Flemina | Inotropin | Inotropisa | Zetarina;
- (MY) Malaysia: Dopaine | Dopamine | Dopmin;
- (NL) Netherlands: Dopamine | Dopaminehydrochloride Eureco Pharma | Dynatra;
- (NO) Norway: Abbodop | Dopamin;
- (NZ) New Zealand: Dopamine | Sterile dopamine concentrate;
- (PA) Panama: Inotropisa;
- (PE) Peru: Dopamina | Dopamina clorhidrato | Intropin;
- (PH) Philippines: Abbott dopamine hcl | Dopabas | Dopamax | Dopaqard | Dopnax | Doprina | Myocard | Myotil;
- (PK) Pakistan: Dopamine | Dopasid | Dopna | Tropin;
- (PL) Poland: Dopamin Giulini | Dopaminum hydrochloricum;
- (PR) Puerto Rico: Dopamine HCL;
- (PT) Portugal: Cordodopa | Cordodopa Forte | Dopamina | Dopamina basi | Medopa;
- (PY) Paraguay: Dopamina duncan | Dopamina northia;
- (QA) Qatar: Dopadren | Dopasel | Myotil;
- (RO) Romania: Clorhidrat de dopamina;
- (RU) Russian Federation: Dofamin | Dofamine | Dofamine Ferein | Dofamine n.s. | Dopamin | Dopamin Admeda | Dopamin Solvay | Dopamine | Dopmin;
- (SA) Saudi Arabia: Domine | Dopamine | Dopamine HCL | Intropin;
- (SE) Sweden: Giludop;
- (SG) Singapore: Dopamine;
- (SI) Slovenia: Dopamin | Dopamin Admeda;
- (SK) Slovakia: Dopamine;
- (SR) Suriname: Dopamine;
- (TH) Thailand: Domin | Dopamex | Dopamin hausmann | Dopamine | Dopamine abbott | Dopamine HCL | Dopamine hydrochloride dbl | Dopaminex | Dopin | Dopmin | Inopin | Upamine;
- (TN) Tunisia: Dopamine | Dopamine mylan;
- (TR) Turkey: Dopadren | Dopamine | Dopamine fresenius | Dopasel | Dopmin | Giludop;
- (TW) Taiwan: Dopamin | Dopamin Giulini | Dopamine | Dopamine HCL | Dopavate | Dopmin | Easydopa | Gipamine | Intropin | Revivan | Uramin | Vidopa;
- (UA) Ukraine: Dofamin darnitsa | Dopamin | Dopamin Admeda | Dopmin;
- (UG) Uganda: Dopamine hcl fresenius;
- (UY) Uruguay: Bagotropin | Dopamin | Dopamina | Dopamina clorhidrato | Dopamina fu | Megadose;

- (VE) Venezuela, Bolivarian Republic of: Dopamina | Dopina | Myotil;
- (ZA) South Africa: Dopamine | Dopamine hcl fresenius | Micro Dopamine;
- (ZM) Zambia: Pharma q dopamine;
- (ZW) Zimbabwe: Pharma q dopamine

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REFERENCES

1. Adams C, Tucker C, Allen B, et al. Disparities in hemodynamic resuscitation of the obese critically ill septic shock patient. *J Crit Care*. 2017;37:219-223. doi:10.1016/j.jcrc.2016.10.004 [PubMed 27969574]
2. American College of Obstetricians and Gynecologists (ACOG) Committee on Practice Bulletins—Obstetrics. Practice Bulletin No. 211: Critical care in pregnancy. *Obstet Gynecol*. 2019;133(5):e303-e319. doi:10.1097/AOG.0000000000003241 [PubMed 31022122]
3. American Heart Association (AHA). CPR & First Aid: Emergency Cardiovascular Care. Adult Bradycardia Algorithm. <https://cpr.heart.org/en/resuscitation-science/cpr-and-ecc-guidelines/algorithms>. Published 2020. Accessed January 29, 2021.
4. American Society of Health-System Pharmacists (ASHP). Pediatric continuous infusion standards. <https://www.ashp.org/-/media/assets/pharmacy-practice/s4s/docs/Pediatric-Infusion-Standards.ashx>. Updated March 2024. Accessed March 22, 2024.
5. Arabi YM, Dara SI, Tamim HM, et al; Cooperative Antimicrobial Therapy of Septic Shock (CATSS) Database Research Group. Clinical characteristics, sepsis interventions and outcomes in the obese patients with septic shock: an international multicenter cohort study. *Crit Care*. 2013;17(2):R72. doi:10.1186/cc12680 [PubMed 23594407]
6. Argalious M, Motta P, Khandwala F, et al. "Renal dose" dopamine is associated with the risk of new-onset atrial fibrillation after cardiac surgery. *Crit Care Med*. 2005;33(6):1327-1332. [PubMed 15942351]
7. Avni T, Lador A, Lev S, Leibovici L, Paul M, Grossman A. Vasopressors for the treatment of septic shock: systematic review and meta-analysis. *PLoS One*. 2015;10(8):e0129305. doi:10.1371/journal.pone.0129305 [PubMed 26237037]
8. Bellomo R, Chapman M, Finfer S, et al. Low-dose Dopamine in Patients With Early Renal Dysfunction: A Placebo-Controlled Randomised Trial. Australian and New Zealand Intensive Care Society (ANZICS) Clinical Trials Group. *Lancet*. 2000;356(9248):2139-2143. [PubMed 11191541]
9. Callaway CW, Donnino MW, Fink EL, et al. Part 8: Post-cardiac arrest care: 2015 American Heart Association guidelines update for cardiopulmonary resuscitation and emergency cardiovascular care [published correction appears in *Circulation*. 2017 Sep 5;136(10):e197]. *Circulation*. 2015;132(18 suppl 2):S465-S482. doi:10.1161/CIR.0000000000000262 [PubMed 26472996]
10. Cardenas-Garcia J, Schaub KF, Belchikov YG, Narasimhan M, Koenig SJ, Mayo PH. Safety of peripheral intravenous administration of vasoactive medication. *J Hosp Med*. 2015;10(9):581-5. doi:10.1002/jhm.2394 [PubMed 26014852]
11. Chan TY. Low-Dose Dopamine in Severe Right Heart Failure and Chronic Obstructive Pulmonary Disease. *Ann Pharmacother*. 1995;29(5):493-496. [PubMed 7655133]
12. Chen HH, Anstrom KJ, Givertz MM, et al. Low-dose dopamine or low-dose nesiritide in acute heart failure with renal dysfunction: the ROSE acute heart failure randomized trial. *JAMA*. 2013;310(23):2533-2543. [PubMed 24247300]
13. Davis AL, Carcillo JA, Aneja RK, et al. American College of Critical Care Medicine (ACCM) clinical practice parameters for hemodynamic support of pediatric and neonatal septic shock. *Crit Care Med*. 2017;45(6):1061-1093. doi:10.1097/CCM.0000000000002425 [PubMed 28509730]

14. De Backer D, Aldecoa C, Njimi H, et al. Dopamine versus Norepinephrine in the Treatment of Septic Shock: A Meta-Analysis. *Crit Care Med*. 2012;40(3):725-730. [PubMed 22036860]
15. De Backer D, Biston P, Devriendt J, et al. Comparison of dopamine and norepinephrine in the treatment of shock. *N Engl J Med*. 2010;362(9):779-789. [PubMed 20200382]
16. Dellinger RP, Levy MM, Rhodes A, et al. Surviving sepsis campaign: international guidelines for management of severe sepsis and septic shock: 2012. *Crit Care Med*. 2013;41(2):580-637. doi:10.1097/CCM.0b013e31827e83af [PubMed 23353941]
17. Dempsey E, Rabe H. The use of cardiotonic drugs in neonates. *Clin Perinatol*. 2019;46(2):273-290. doi:10.1016/j.clp.2019.02.010 [PubMed 31010560]
18. Denkler KA, Cohen BE. Reversal of dopamine extravasation injury with topical nitroglycerin ointment. *Plast Reconstr Surg*. 1989;84(5):811-813. [PubMed 2510208]
19. Domonoske C. Appendix A: Common neonatal intensive care unit (NICU) medication guidelines. In: Eichenwald EC, Hansen AR, Martin CR, Stark AR. *Cloherty and Stark's Manual of Neonatal Care*. 8th ed. Lippincott Williams & Wilkins; 2017.
20. Dopamine HCL injection solution [prescribing information]. Shirley, NY: American Regent Inc; March 2014.
21. Dopamine Hydrochloride Injection (Fliptop, Pintop, LifeShield Fliptop Vials) [prescribing information]. Lake Forest, IL: Hospira; September 2023.
22. Dopamine Hydrochloride in 5% Dextrose Injection [prescribing information]. Lake Forest, IL: Hospira Inc; A 2025.
23. Elkayam U, Ng TM, Hatamizadeh P, Janmohamed M, Mehra A. Renal Vasodilatory Action of Dopamine in Patients With Heart Failure: Magnitude of Effect and Site of Action. *Circulation*. 2008;117(2):200-205. *J Card Fail*. 2010;16(12):922-930. [PubMed 18172028]
24. Erstad BL, Barletta JF. Drug dosing in the critically ill obese patient: a focus on medications for hemodynamic support and prophylaxis. *Crit Care*. 2021;25(1):77. doi:10.1186/s13054-021-03495-8 [PubMed 33622380]
25. Evans L, Rhodes A, Alhazzani W, et al. Surviving Sepsis Campaign: international guidelines for management of sepsis and septic shock 2021. *Crit Care Med*. 2021;49(11):e1063-e1143. doi:10.1097/CCM.0000000000005337 [PubMed 34605781]
26. Expert opinion. Senior Obesity Editorial Team: Jeffrey F. Barletta, PharmD, FCCM; Manjunath P. Pai, PharmD, FCP; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC.
27. Friedrich JO, Adhikari N, Herridge MS, Beyene J. Meta-analysis: low-dose dopamine increases urine output but does not prevent renal dysfunction or death. *Ann Intern Med*. 2005;142(7):510-524. [PubMed 15809463]
28. Heidenreich PA, Bozkurt B, Aguilar D, et al. 2022 AHA/ACC/HFSA guideline for the management of heart failure: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation*. 2022;145(18):e895-e1032. doi:10.1161/CIR.0000000000001063 [PubMed 35363499]
29. Hollenberg SM, Ahrens TS, Annane D, et al. Practice Parameters for Hemodynamic Support of Sepsis in Adult Patients: 2004 Update. *Crit Care Med*. 2004;32(9):1928-1948. [PubMed 15343024]
30. Hollenberg SM. Vasoactive drugs in circulatory shock. *Am J Respir Crit Care Med*. 2011;183(7):847-855. doi:10.1164/rccm.201006-0972CI [PubMed 21097695]
31. Jeejeebhoy FM, Zelop CM, Lipman S, et al; American Heart Association Emergency Cardiovascular Care Committee, Council on Cardiopulmonary, Critical Care, Perioperative and Resuscitation, Council on Cardiovascular Diseases in the Young, Council on Clinical Cardiology. Cardiac arrest in pregnancy: a scientific statement from the American Heart Association. *Circulation*. 2015;132(18):1747-1773. doi:10.1161/CIR.0000000000000300 [PubMed 26443610]

32. Johnson RL Jr. Low-Dose Dopamine and Oxygen Transport by the Lung. *Circulation*. 1998;98(2):97-99. [PubMed 9679713]
33. Kellum JA, Decker J. Use of Dopamine in Acute Renal Failure: A Meta-Analysis. *Crit Care Med*. 2001;29(8):1526-1531. [PubMed 11505120]
34. Kidney Disease: Improving Global Outcomes (KDIGO) Acute Kidney Injury Work Group. KDIGO clinical practice guideline for acute kidney injury. *Kidney Int*. 2012;2(suppl 1):1-138. <https://kdigo.org/wp-content/uploads/2016/10/KDIGO-2012-AKI-Guideline-English.pdf>.
35. Kleinman ME, Chameides L, Schexnayder SM, et al. Part 14: pediatric advanced life support: 2010 American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care. *Circulation*. 2010;122(18)(suppl 3):S876-S908. [PubMed 20956230]
36. Kusumoto FM, Schoenfeld MH, Barrett C, et al. 2018 ACC/AHA/HRS guideline on the evaluation and management of patients with bradycardia and cardiac conduction delay: a report of the American College of Cardiology/American Heart Association task force on clinical practice guidelines and the Heart Rhythm Society. *J Am Coll Cardiol*. 2019;74(7):e51-e156. doi:10.1016/j.jacc.2018.10.044 [PubMed 30412709]
37. Lamontagne F, Richards-Belle A, Thomas K, et al; 65 trial investigators. Effect of reduced exposure to vasopressors on 90-day mortality in older critically ill patients with vasodilatory hypotension: a randomized clinical trial. *JAMA*. 2020;323(10):938-49. doi:10.1001/jama.2020.0930 [PubMed 32049269]
38. Levy MM, Evans LE, Rhodes A. The Surviving Sepsis Campaign Bundle: 2018 Update. *Crit Care Med*. 2018;46(6):997-1000. doi: 10.1097/CCM.0000000000003119 [PubMed 29767636]
39. Lewis T, Merchan C, Altshuler D, Papadopoulos J. Safety of the peripheral administration of vasopressor agents. *J Intensive Care Med*. 2019;34(1):26-33. doi:10.1177/0885066616686035 [PubMed 28073314]
40. Manaker S, Teja B. Use of vasopressors and inotropes. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed November 22, 2021.
41. Martin C, Papazian L, Perrin G, et al. Norepinephrine or Dopamine for the Treatment of Hyperdynamic Septic Shock? *Chest*. 1993;103(6):1826-1831. [PubMed 8404107]
42. Medlej K, Kazzi AA, El Hajj Chehade A, et al. Complications from administration of vasopressors through peripheral venous catheters: an observational study. *J Emerg Med*. 2018;54(1):47-53. doi:10.1016/j.jemermed.2017.09.007 [PubMed 29110979]
43. Murray KL, Wright D, Laxton B, Miller KM, Meyers J, Englebright J. Implementation of standardized pediatric i.v. medication concentrations. *Am J Health Syst Pharm*. 2014;71(17):1500-1508. [PubMed 25147175]
44. Nemecek BD, Hammond DA, eds. *Demystifying Drug Dosing in Renal Dysfunction*. American Society of Health-System Pharmacists; 2019.
45. Panchal AR, Bartos JA, Cabañas JG, et al; Adult Basic and Advanced Life Support Writing Group. Part 3: adult basic and advanced life support: 2020 American Heart Association guidelines for cardiopulmonary resuscitation and emergency cardiovascular care. *Circulation*. 2020;142(16)(suppl 2):S366-S468. doi:10.1161/CIR.0000000000000916 [PubMed 33081529]
46. Patel GP, Grahe JS, Sperry M, et al. Efficacy and Safety of Dopamine versus Norepinephrine in the Management of Septic Shock. *Shock*. 2010;33(4):375-380. [PubMed 19851126]
47. Peberdy MA, Callaway CW, Neumar RW, et al. Part 9: Post Cardiac Arrest Care: 2010 American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care. *Circulation*. 2010;122(18)(suppl 3):768-786. [PubMed 20956225]
48. Phillips MS. Standardizing I.V. Infusion Concentrations: National Survey Results. *Am J Health Syst Pharm*. 2011;68(22):2176-2182. [PubMed 22058104]

49. Prasanna N, Yamane D, Haridasa N, Davison D, Sparks A, Hawkins K. Safety and efficacy of vasopressor administration through midline catheters. *J Crit Care*. 2021;61:1-4. doi:10.1016/j.jcrc.2020.09.024 [PubMed 33049486]
50. Prescott HC, Antonelli M, Alhazzani W, et al. Surviving sepsis campaign: international guidelines for management of sepsis and septic shock 2026. *Crit Care Med*. 2026;54(4):725-812. doi:10.1097/CCM.0000000000007075 [PubMed 41869847]
51. Radosevich JJ, Patanwala AE, Erstad BL. Norepinephrine dosing in obese and nonobese patients with septic shock. *Am J Crit Care*. 2016;25(1):27-32. doi:10.4037/ajcc2016667 [PubMed 26724290]
52. Refer to manufacturer's labeling.
53. Reynolds PM, Maclaren R, Mueller SW, Fish DN, Kiser TH. Management of extravasation injuries: a focused evaluation of noncytotoxic medications. *Pharmacotherapy*. 2014;34(6):617-632. [PubMed 24420913]
54. Rhodes A, Evans LE, Alhazzani W, et al. Surviving Sepsis Campaign: International Guidelines for Management of Sepsis and Septic Shock: 2016. *Intensive Care Med*. 2017;43(3):304-377. doi:10.1007/s00134-017-4683-6 [PubMed 28101605]
55. Russell JA, Gordon AC, Williams MD, Boyd JH, Walley KR, Kissoon N. Vasopressor therapy in the intensive care unit. *Semin Respir Crit Care Med*. 2021;42(1):59-77. doi:10.1055/s-0040-1710320 [PubMed 32820475]
56. Sakr Y, Reinhart K, Vincent JL, et al. Does Dopamine Administration in Shock Influence Outcome? Results of the Sepsis Occurrence in Acutely Ill Patients (SOAP) Study. *Crit Care Med*. 2006;34(3):589-597. [PubMed 16505643]
57. Shenoi RP, Timm N; Committee on Drugs; Committee on Pediatric Emergency Medicine. Drugs used to treat pediatric emergencies. *Pediatrics*. 2020;145(1):e20193450. [PubMed 31871244]
58. Sinha SS, Morrow DA, Kapur NK, Kataria R, Roswell RO. 2025 concise clinical guidance: an ACC expert consensus statement on the evaluation and management of cardiogenic shock: a report of the American College of Cardiology Solution Set Oversight Committee. *J Am Coll Cardiol*. 2025;85(16):1618-1641. doi:10.1016/j.jacc.2025.02.018 [PubMed 40100174]
59. Stefanos SS, Kiser TH, MacLaren R, Mueller SW, Reynolds PM. Management of noncytotoxic extravasation injuries: a focused update on medications, treatment strategies, and peripheral administration of vasopressors and hypertonic saline. *Pharmacotherapy*. 2023;43(4):321-337. doi:10.1002/phar.2794 [PubMed 36938775]
60. Stier PA, Bogner MP, Webster K, et al. Use of Subcutaneous Terbutaline to Reverse Peripheral Ischemia. *Am J Emerg Med*. 1999;17(1):91-94. [PubMed 9928712]
61. Tian DH, Smyth C, Keijzers G, et al. Safety of peripheral administration of vasopressor medications: a systematic review. *Emerg Med Australas*. 2020;32(2):220-227. doi:10.1111/1742-6723.13406 [PubMed 31698544]
62. Tisdale JE, Patel R, Webb CR, Borzak S, Zarowitz BJ. Electrophysiologic and proarrhythmic effects of intravenous inotropic agents. *Prog Cardiovasc Dis*. 1995;38(2):167-180. [PubMed 7568905]
63. Triposkiadis FK, Butler J, Karayannis G, et al. Efficacy and safety of high dose versus low dose furosemide with or without dopamine infusion: The Dopamine in Acute Decompensated Heart Failure II (DAD-HF II) trial. *Int J Cardiol*. 2014;172(1):115-121. [PubMed 24485633]
64. US Food and Drug Administration; Institute for Safe Medication Practices. FDA and ISMP Lists of Look-Alike Drug Names With Recommended Tall Man (Mixed Case) Letters. ISMP; 2023.
65. Vadiiei N, Daley MJ, Murthy MS, Shuman CS. Impact of norepinephrine weight-based dosing compared with non-weight-based dosing in achieving time to goal mean arterial pressure in obese patients with septic shock. *Ann Pharmacother*. 2017;51(3):194-202. doi:10.1177/1060028016682030 [PubMed 27886982]

66. van de Borne P, Oren R, Somers VK. Dopamine Depresses Minute Ventilation in Patients With Heart Failure. *Circulation*. 1998;98(2):126-131. [PubMed 9679718]
67. van Diepen S, Katz JN, Albert NM, et al; American Heart Association Council on Clinical Cardiology; Council on Cardiovascular and Stroke Nursing; Council on Quality of Care and Outcomes Research; Mission: Lifeline. Contemporary management of cardiogenic shock: a scientific statement from the American Heart Association. *Circulation*. 2017;136(16):e232-e268. doi:10.1161/CIR.0000000000000525 [PubMed 28923988]
68. Vasu TS, Cavallazzi R, Hirani A, et al. Norepinephrine or Dopamine for Septic Shock: Systematic Review of Randomized Clinical Trials. *J Intensive Care Med*. 2012;27(3):172-178. [PubMed 21436167]
69. Vermont Oxford Network (VON). Neonatal drug concentrations. Updated November 2022. Accessed March 22, 2024. <https://public.vtoxford.org/wp-content/uploads/2022/11/Neonatal-Drug-Concentrations-Updated-November-2022.pdf>
70. Wan SH, Stevens SR, Borlaug BA, et al. Differential response to low-dose dopamine or low-dose nesiritide in acute heart failure with reduced or preserved ejection fraction: results from the ROSE AHF trial (Renal Optimization Strategies Evaluation in Acute Heart Failure). *Circ Heart Fail*. 2016;9(8):e002593. doi:10.1161/CIRCHEARTFAILURE.115.002593 [PubMed 27512103]
71. Weiss SL, Peters MJ, Alhazzani W, et al. Surviving Sepsis Campaign international guidelines for the management of septic shock and sepsis-associated organ dysfunction in children. *Pediatr Crit Care Med*. 2020;21(2):e52-e106. doi:10.1097/PCC.0000000000002198 [PubMed 32032273]
72. Wigginton JG, Agarwal S, Bartos JA, et al. Part 9: adult advanced life support: 2025 American Heart Association guidelines for cardiopulmonary resuscitation and emergency cardiovascular care. *Circulation*. 2025;152(16_suppl_2):S538-S577. doi:10.1161/CIR.0000000000001376 [PubMed 41122884]
73. Wong AF, McCulloch LM, Sola A. Treatment of Peripheral Tissue Ischemia With Topical Nitroglycerin Ointment in Neonates. *J Pediatr*. 1992;121(6):980-983. [PubMed 1447671]
74. Young JB, Moen EK. Outpatient parenteral inotropic therapy for advanced heart failure. *J Heart Lung Transplant*. 2000;19(8)(suppl):s49-s57. [PubMed 11016488]
75. Zipes DP, Camm AJ, Borggrefe M, et al; American College of Cardiology/American Heart Association Task Force; European Society of Cardiology Committee for Practice Guidelines; European Heart Rhythm Association; Heart Rhythm Society. ACC/AHA/ESC 2006 Guidelines for Management of Patients With Ventricular Arrhythmias and the Prevention of Sudden Cardiac Death: a report of the American College of Cardiology/American Heart Association Task Force and the European Society of Cardiology Committee for Practice Guidelines (writing committee to develop Guidelines for Management of Patients With Ventricular Arrhythmias and the Prevention of Sudden Cardiac Death): developed in collaboration with the European Heart Rhythm Association and the Heart Rhythm Society. *Circulation*. 2006;114(10):e385-e484. [PubMed 16935995]

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